Toward a qSIM-Based PQC-QKD Hybrid Secure Tunnel for MUM-T*

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Abstract

MUM-T environments require secure communication between unmanned platforms and command-and-control systems. However, these links are exposed to real-time eavesdropping and Harvest-Now-Decrypt-Later threats, while direct QKD deployment on unmanned vehicles remains challenging due to weight, cost, and operational constraints. This paper proposes Quantum Secure Tunnel (QST), which pre-injects QKD-derived PSKs into a KCMVP Security Level 2 cryptographic module (qSIM) and establishes a TLS 1.3 psk_dhe_ke tunnel combining the preloaded PSK with X25519MLKEM768, conveying the key_id via the standard pre_shared_key identity field without protocol modification. A Raspberry Pi prototype verifies qSIM PSK loading, QKMS recovery, and TLS 1.3 tunnel establishment.

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1 Introduction and Related Work

MUM-T (Manned-Unmanned Teaming) is a tactical configuration in which unmanned platforms operate alongside manned command-and-control (C2) systems. Military data over the vehicle-to-C2 link is simultaneously exposed to real-time eavesdropping and Harvest-Now-Decrypt-Later (HNDL) threats, demanding quantum security technologies such as Quantum Key Distribution (QKD), Post-Quantum Cryptography (PQC), and quantum cryptographic modules (QCM). Among these, QKD cannot be deployed on unmanned vehicles due to operational constraints, creating an end-to-end quantum security gap — the QKD last-mile problem addressed in this paper.

We propose and implement QST (Quantum Secure Tunnel), which pre-injects QKD PSKs into a KCMVP Security Level 2 cryptographic module (QCM) as a quantum SIM (qSIM). QST establishes an end-to-end TLS 1.3 tunnel by combining the preloaded QKD PSK with X25519MLKEM768 hybrid key exchange [1].

2 Quantum Secure Tunnel System Design

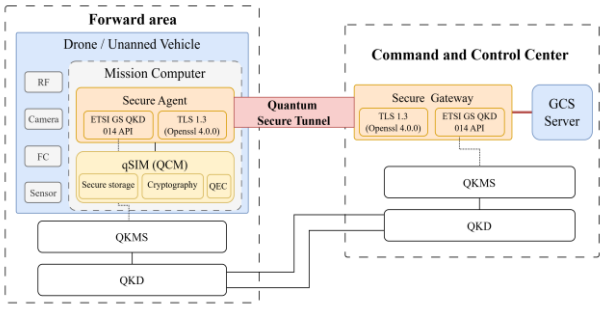


Figure 1: QST architecture for MUM-T end-to-end security

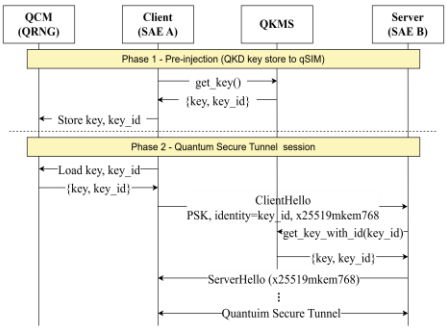


Figure 2: QST protocol sequence

Fig. 1 illustrates the end-to-end architecture of QST. As shown in Fig. 2, Phase 1 (pre-injection) stores QKD-derived keys in the vehicle's qSIM before mission deployment; during the mission, the preloaded key material is used to establish the QST tunnel. To address the last-mile problem of vehicles unable to connect to QKD infrastructure, we propose using a QCM as a qSIM (Quantum SIM). Like a mobile network SIM card, it stores the military or adopting organization's keys, certificates, and credentials within a logically and physically protected module.

3 Implementation, Results & Conclusion

We implemented the QST prototype on two Raspberry Pi 4 (aarch64) devices running OpenSSL 4.0.0 with native X25519MLKEM768 support. The client inserted the qSIM-loaded key_id in the PskIdentity.identity field and initiated a TLS 1.3 psk_dhe_ke handshake; the server called the QKMS get_key_with_id API using the received key_id to recover the PSK and established the QST session.

Among related works, QKD-KEM [2] and Frauenschläger et al. [3] combine PQC-KEM with QKD within the scope of existing RFC standards. However, both assume fixed network infrastructure and require library modifications, leaving unmanned platform deployment and hardware key storage unaddressed. QST addresses these limitations by supporting PQC-KEM integration, QKD integration, unmanned platform deployment, hardware key storage, and RFC-compliant operation without non-

standard TLS extensions. Future work includes OpenVPN integration, ML-DSA fallback hardening, and KpqC algorithm support.

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